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ARTICLE

Inclusive Deaf Education Enabled by Artificial Intelligence: The Path to a Solution

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Abstract

Deaf learners in the Global South struggle to access equitable education, in particular, there are few instances where they can be facilitated in inclusive classrooms. The challenges include a lack of teachers that can sign proficiently, the unavailability of interpreters and few teachers trained in Deaf education. This paper explores the potential and challenges of an approach that aims to facilitate real-time communication between Deaf students and non-signing teachers, by combining speech and language technologies, computer vision, machine translation and 3-D avatars, all underpinned by artificial intelligence. A closed-loop solution is proposed that takes advantage advances in gesture recognition, machine translation and speech and language technologies to allow for communication between Deaf students and hearing teachers, both using their preferred medium of communication. A focus group was held with members of the Deaf community to gauge their interest and concerns about the proposed system. The results revealed a fair level of curiosity and concern. Five major themes emerged: Avatar Effectiveness and Realism; Language and Cultural Considerations; Resource and Cost Concerns; Real-time Translation Challenges and Community Involvement and Feedback. The results revealed important issues concerning bilingual-bicultural access for the hearing teacher and the need for sign language production systems to factor in the nuances of unfamiliar concepts, essentially emulating the role of Deaf Relay Interpreters.

Keywords AI-enabled Inclusive Deaf Education · AI in Education · AI for Marginalised Populations · AI in the Global South · AI for Education in the Global South

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Introduction

Worldwide, it is estimated that 430 million people have hearing loss requiring rehabilitation (World Health Organisation, 2024). Factors such as hearing thresholds, type and severity of hearing loss, age of onset of hearing loss, family makeup and socio-cultural background, shape the Deaf[1] individual and impact their access to education (Guardino et al., 2018; Domagała-Zyśk, 2013; Humphries et al., 2012; Knoors & Marschark, 2012). Approximately 80% of the Deaf population, worldwide, do not have access to any education (World Federation of the Deaf, 2016); this figure is heavily influenced by the fact that 80% of Deaf persons live in low- and middle-income countries, where access to education is not equitable (World Health Organisation, 2024). This problem carries forward, as many struggle to find meaningful employment (O'Connell, 2023; Gupta et al., 2023); in one extreme case, it has been shown that approximately 75% of secondary-level Deaf graduates have no marketable skills by school leaving age (Planning Institute of Jamaica, 2016). This leads to the unfortunate circumstance, where a significant proportion of the Deaf are unemployed or underemployed (Dammeyer, 2019; Garberoglio et al., 2019, Todorov, 2022). This is not surprising, as the lack of formal education, pointed to above, must surely negatively impact employability. Even where attempts are made to provide education, accessibility of the education offered is a challenge (Straaten et al., 2021). There are many reasons for this, but the most significant are a lack of teachers trained in Deaf education, interpreters, assistive devices and other tools to facilitate accessibility (Lee & Low, 2013; Leyser et al., 2011; Sharma & Das, 2015; Wapling, 2016; Kelly et al., 2022b). Where they do have access to education, the options for the Deaf are either to be educated at home, in specialist schools for the Deaf, inclusion in mainstream schools with no additional support, or inclusion in mainstream schools with specialist support (Coy, 2022; Silvestri & Hartman, 2022).

In the Global North (the so-called, developed world), inclusive education often means that Deaf children attend mainstream schools where they are supported by a wide variety of technological devices and human assistance, including interpreters and teachers who sign (Humphries et al., 2012; Knoors & Marschark, 2012). This ecosystem is important for the Deaf to make the most of their opportunity to gain an education. Despite the investment in resources to support Deaf education, it is well documented that Deaf students generally have lower levels of educational attainment (Järvelin et al., 1997; Richardson, 2001; Rydberg et al., 2009; Sharma et al., 2023; Muller et al., 2023). This includes the Deaf in rich nations, such as the United States, The United Kingdom, Sweden and Finland (Garberoglio, 2021; Marschark et al., 2015; Hutchinson, 2023; Antonson et al., 2006; Rydberg et al., 2009; Järvelin et al., 1997). A recent large, longitudinal, retrospective study of primary school student achievement in the Netherlands, between 2008 and 2018, showed that Deaf students had lower scores in Mathematics and Language than their hearing counterparts, even when the Deaf students attended mainstream schools (van der Straaten et al., 2021). The Deaf attending mainstream elementary schools (approximately two-thirds of the

[1] Deaf is used to mean a person with hearing loss significant enough to negatively impact their education (Jarvis et al., 2002).

Deaf students in the study) went on to the same secondary schools as their hearing counterparts, despite having lower scores. According to the study, the Deaf students attending specialised schools were far more likely to attend lower-level post-primary educational institutions and rarely went on to post-secondary education or vocational training.

While it is widely reported that Deaf students in mainstream education are at a significant advantage to their counterparts in specialist schools, questions remain about the suitability of this approach, even in the developed economies in which they are prevalent (Kreimeyer, 2000; Silvestri & Hartman, 2022). Owing to the challenges with background noise, low level conversations and challenges following the instructions of teachers, "instructions in mainstream settings are not always communicatively accessible" to the Deaf (van der Straaten et al., 2021, p. 407). Inclusion in mainstream classrooms is largely facilitated by the use of cochlear implants (CI) by deaf students (Muller et al., 2023). Notwithstanding this drive, Deaf students with CI's have a number of challenges related to their use including challenges in acquiring the nuances of spoken language and a lack of sign language; this leaves them at a disadvantage academically and socially, as they do not have a natural place in either the hearing or the Deaf communities (Archbold, 2015; Knoors & Marschark, 2015; Hilviu et al., 2021; Šantić & Bonetti, 2023). Wearers of hearing aids are also at a disadvantage in mainstream education, as they are often thought to be performing relatively well, compared to CI users, and are thus offered less support (Marschark et al., 2015). In many countries it has been recognised that Deaf students, even those with some degree of oral language, benefit academically and socially from a bilingual-bicultural approach, which incorporates sign language and Deaf culture into mainstream education (Skerrit, 2020; Dammeyer & Marschark, 2016). This approach better aligns with the ideal inclusion model as it meets the needs of the Deaf rather than requiring them to adapt to the provisions available to the regular stream of students in the system. Despite the benefits of this approach, it has not been widely adopted (World Federation of the Deaf, 2016) owing to disagreements about the most suitable approach to educating Deaf individuals and concerns about the potential harm that bilingualism can cause to the Deaf individual's ability to acquire spoken language (Jones, 2021; Pontecorvo, 2023). The challenge is made even more significant, as less than 4% of children acquire sign language from birth as a result of exposure to Deaf parents who sign (Mitchell & Karchmer, 2004). The implication of this is that over 95% of Deaf/hard of hearing (D/hh) children are born to parents who are hearing and whose primary means of communication is a spoken language (Archbold, 2015; Terry, 2023). These parents face the daunting challenge of being unprepared for the birth of the D/hh child, which includes the lack of proficiency in a language that is accessible to their child (Humphries et al., 2019).

The bilingual-bicultural model of inclusive education may not be feasible in the developing world, as there are challenges that are unique to the developing world given their smaller economies, approach to education and attitudes to the Deaf. There are questions about the acceptability of this approach, given that there are technical, financial and socio-cultural challenges that attend this approach to equitable education (Cooper, 2019; Punch & Hyde, 2011). Some of the issues include the cost of assistive devices, maintenance and rehabilitation costs as well as a shortage of trained

surgeons to implant CIs and audiologists to support the Deaf in their use (Magro et al., 2018; Jesuyajolu et al., 2023). It must be noted that some studies have shown that CIs are cost-effective in low- and middle-income countries (Emmett et al., 2016, 2019; Emmett et al., 2015). However, that does not appear to be the reality on the ground, as, globally, less than 1% of people who could potentially benefit from the use of a CI have access to one (Bodington et al., 2021). Another reason why the bilingual-bicultural model may not work in the Global South is that the typical model of inclusive education is predicated on the assumption that the student has spoken language, or at least, the ability to read the main language of their country, which is not always the case for the Deaf (Counselman, Meltzer & Marquart, 2020; Tang, 2024; Alanazi, 2021; Gagnon et al., 2021; Tsach & Most, 2016). This is because reading proficiency is achieved through phonological awareness, phonics, word recognition, vocabulary acquisition and fluency, skills that are not accessible to many Deaf persons (Hartman et al., 2019; Runnion & Gray, 2019; Wass et al., 2019). It should be noted however, that some profoundly Deaf learners have been able to develop their reading skills through the acquisition of sign language as their first language (Verlinden et al., 2005; Howerton-Fox & Falk, 2019).

Even if assistive technologies were widely accessible in the Global South, the bilingual-bicultural approach is not culturally responsive to all Deaf communities, in particular those communities that would prefer to not use a spoken language. Having a culturally appropriate solution to the education of the Deaf is critical, as some within the Deaf community view the use of assistive devices (such as cochlear implants and hearing aids) as an attack on Deaf culture and are openly hostile to the use of the technology and to those who use it (Cooper, 2019; Sparrow, 2010; Coy, 2022). Such an approach must give prominence to the use of sign language in Deaf education, given it is the native language of the Deaf and they should not be forced to rely on spoken language as their only means to achieving equity in education. The challenges for Deaf education are different in the Global North and south (the developing world). However, there is compelling evidence, from the Global North, that allowing Deaf learners to use sign language as they engage in the teaching and learning process, has tremendous benefits for their overall development (Dammeyer & Marschark, 2016; Nielsen, 2016; Goodwin & Lillo-Martin, 2019; Hoffmeister et al., 2022) and that depriving them of an accessible language is certainly harmful (Secora & Smith, 2021; Hall et al., 2019).

In summary, inclusive Deaf education is inadequate in both the Global North and south. In most inclusive classrooms in the Global North, sign language is not always accommodated (Powell et al., 2014; Constantinou et al., 2020) or attempts to incorporate sign language is done through teachers who aren't trained, or with the use of interpreters who are not teachers, which means that students often miss out on important information (Ngobeni et al., 2020; Jones et al., 2021; Kelly et al., 2022b). In the Global South, assistive technologies are not readily available, nor are trained teachers who sign. This leaves a space for the use of technology to fill the gaps evident in the mainstream classrooms of both domains - the need for increased use of sign language in bilingual-bicultural Deaf education in the north and the need for a system for facilitating sign language use in the south, where the vast majority of the Deaf have no access to the language of instruction. While there have been some calls for the

inclusion of technology to enhance access to education for the Deaf (Constantinou et al., 2020; Lynch et al., 2024) - including recent calls for systems incorporating, or developed by, artificial intelligence (Sreemathy et al., 2023; Levesque et al., 2023) - the existing literature lacks a proposal for a comprehensive solution to the myriad challenges outlined.

This paper proposes a novel approach to the provision of education for the Deaf as stakeholders seek to have equitable Deaf education world-wide. The solution is applicable to the Global South, given that the majority of the challenges, and potential users, exist there. Additionally, it has been designed for potential future use, with appropriate modifications, in the Global North, as the challenges are not unique to the developing world. A framework and approach are proposed that employ the extensive use of AI solutions to facilitate the education of the Deaf in mainstream classrooms, even if they have no access to spoken language and their teacher is unable to sign. The solution is envisioned in such a way as to be culturally appropriate and tailored to the Deaf, while not being disruptive to the hearing students or the teacher.

The research is guided by the following research questions:

1. What are the main components needed for a closed-loop system that foster inclusive Deaf education in the Global South?
2. What are the most important concerns of the Deaf community as it relates to the development and use of such a system.

The rest of the paper is structured as follows: Sect. 2 explores the state of Deaf education in both Global North and south contexts, common barriers to inclusive Deaf education and how AI can play a pivotal role in fostering inclusive Deaf education. The guiding theoretical framework is presented in Sect. 3, which is followed by a detailed description of the proposed solution, its value, and potential limitations. A focus group, exploring the reaction of members of the Deaf community in Jamaica to the proposed solution, is described in Sect. 5. The results, analysis and discussion follow and the paper is concluded in Sect. 8.

Issues Impacting Deaf Education

An Overview of Deaf Education

Equity in education for the disabled is a global imperative, which is supported by a variety of international organisations (Barton & Armstrong, 2007; UNESCO, 2005). After several international declarations, including most recently, the Agenda for Sustainable Development education goal (SDG 4), most Deaf students in the Global North are educated in inclusive classrooms. This, however, is not universally accepted as producing the most beneficial outcomes for those in the disabled community (Venail et al., 2010; Sarant et al., 2015; Nunes et al., 2001; Krijger et al., 2020). Nonetheless, the drive to have Deaf students in inclusive education continues unabated because it is believed that inclusion is the pathway to equitable education (UNESCO, 2001; OECD, 2003).

The widespread implantation of the cochlear implant has been credited with bolstering the effort to move, almost exclusively, toward inclusive schooling in the developed world (Archbold, 2015; Marschark et al., 2019; Duarte et al., 2016). A CI is a surgically implanted device that by-passes the normal acoustic pathways to provide the Deaf user with a modified sense of sound; a CI does not help the Deaf to recover lost hearing. Thus, while there are noted benefits to the education of the Deaf, the overall impact is mixed. Early implantation, coupled with speech and language training specific to the access to speech sounds that the CI provides, is shown to confer educational and employment benefits to Deaf students with CIs compared to those without them (Venail et al., 2010; Sarant et al., 2015; Spencer et al., 2012; Huber, 2018; Alasim, 2021). However, there are also challenges, including: difficulty in acquiring the finer points of spoken language, social and a loss of educational support as they grow older (Archbold, 2015; Knoors & Marschark, 2015).

In many developed countries, the majority of Deaf children access inclusive education. They are supported by personal assistive devices, such as hearing aids, induction loops, frequency modulation (FM) systems, and cochlear implants, as well as classroom teachers who sign, interpreters or itinerant teachers of the Deaf (Knoors & Marschark, 2012). The use of assistive technology for Deaf education has become so normalised, only 1–2% of Deaf people across the globe access education through sign language (World Federation of the Deaf, 2016). This is unfortunate, as many Deaf people, particularly those in the developed world, function as bimodal bilinguals; this ability to utilise both oral and sign language is known to impact the way they learn (Knoors & Marschark, 2012). Deaf children need to be exposed to sign language at an early age, given that delayed language acquisition leads to language delay and lower levels of academic attainment (Dammeyer & Marschark, 2016; Magnuson, 2000). A lack of support for bilingualism has implications for relationships with the Deaf community, which can leave the Deaf at risk of isolation and a lack of identity, especially in CI users who may have challenges with oral language (Bat-Chava & Deignan, 2001; Wheeler et al., 2007; Punch & Hyde, 2011).

An understanding of the value of bilingualism has led to support for bilingual-bicultural education in countries such as Sweden, Norway and Denmark (Swanwick, 2014; Dammeyer & Marschark, 2016), the United Kingdom (Powers, 2002), the United States (Humphries, 2013), Hong Kong (Tang, 2017) and Zimbabwe (Sibanda & Tlale, 2019). These programmes include all stakeholders: family, teachers and administrators.

Education of the Deaf in the Global South

The developed world has, generally speaking, subscribed to a model of equity in education for the Deaf referred to as inclusion. This is feasible because of the availability of assistive technology, which is readily available in these contexts. Assistive technologies are not as readily available in the developing world, which has a profound negative impact on the educational and overall life opportunities of the Deaf in developing countries (de Witte et al., 2019; Matter et al., 2017; McPherson, 2014). Thus, even though developing countries, such Brazil, Chile, Jamaica, Nigeria and South Africa (Tanure Alves et al., 2021; Rosas et al., 2023; McArthur, Ewen-Smith &

Scott, 2021; Jesuyajolu et al., 2023; Walton & Engelbrecht, 2022), are moving in that direction, a number of issues, including culture, lack of infrastructure and financial resources to support inclusion act as barriers (Lee & Low, 2013; Leyser et al., 2011; Kisanga, 2022; Ngobeni et al., 2020; Kelly et al., 2022b).

Some developing countries attempt to offer inclusive education. However, what often occurs is simply the application of an operational definition of inclusion achieved by providing the Deaf a place in mainstream schools. Without the required technical and physical infrastructure specialist teachers and learning resources, this approach is of little value to the Deaf (Singal, 2008; Ngcobo & Muthukrishna, 2011; Sharma & Das, 2015; Wapling, 2016). Often, the Deaf are not well catered for in mainstream schools, in part because they are not well understood, teachers do not have specialist training, and sign language is not the language of instruction (Armstrong, 2005; Kelly et al., 2022).

Barriers to Inclusive Deaf Education in the Global South

While countries have the best of intentions as it concerns equitable education, there is not always the effort required to follow through on this promise. Budgetary allocations for educating the Deaf have been cut in developing and developed countries (Mertzani, 2015; National Deaf Children's Society, 2019; Strauss, 2019). This limits what can be done to provide the required accommodations for the Deaf learner, whether in a mainstream school or a Deaf only specialist school.

Without a coherent policy and overarching legislative framework covering Deaf education in many developing countries, there is a significant impediment to equitable education (Mutasa, 2000). Without a systematic effort to provide support for the Deaf learner, all attempts to move to inclusive education will end in failure and be to the detriment of the Deaf (Cheong et al., 2019). Barriers exist, even where resources are available, they may not be accessible. The cost of assistive technology is either too high, skilled doctors are not available, or the Deaf community is unaware of the availability or benefits of assistive devices (McPherson, 2014).

Inaccessibility is expressed in other ways also. At times, projectors, and tablets, are provided to support Deaf education. For the Deaf student without assistive devices, reading from a projector, lip-reading, and working with an interpreter may cause cognitive overload, given the attempt to integrate the information from all three sources (Domagała-Zyśk, 2013). Much of the information displayed using these visual aids is presented as text. This assumes that the Deaf student is fluent in the language of instruction, which is not always the case.

The Deaf have historically been seen as being disabled and without prospects for a productive life (Hensley, 2015). Often, they are treated as though they need to be cured of a terrible affliction (Humphries et al., 2012; Sparrow, 2005). There is a significant backlash against those attitudes, as many Deaf persons are quite proud of their Deafness. These members of the community view Deafness as part of their cultural identity and sign language as a linguistic marker of this identity (Chapman & Dammeyer, 2017; Foley & Beth, 2012; Goggin et al., 2003). To these individuals, the implantation of cochlear implants in young Deaf children is seen as a form of cultural erasure or ethnocide (Sparrow, 2010). As a result of these and other concerns,

many culturally Deaf individuals refuse to use assistive devices and reject the model of inclusive education championed in the developed world. They prefer, instead, a model that allows them to express themselves culturally, where sign language is the language of instruction (Chapman & Dammeyer, 2017). This antagonism has led to some members of the community ostracising users of assistive devices, often because they are not fluent in sign language and are perceived to have rejected Deaf culture (Leigh, 2009).

A significant barrier to inclusive education for the Deaf is the lack of trained teachers of the Deaf or even trained teachers who sign (Gamal, 2022; De Meulder & Haualand, 2021). Where teachers do not have these specialist skills, they believe that teaching a Deaf child is difficult and should only be done by those with the particular skill sets (Andrews et al., 2021; Walton & Engelbrecht, 2022). A number of solutions have been proposed to deal with the shortage of teachers who sign, including: training more teachers to sign (World Federation of the Deaf, 2018, Alawajee, 2022), providing support for in-service teachers (Kelly et al., 2022a, b) and hiring more sign language interpreters to support learning (De Meulder & Haualand, 2021). Each approach has its own challenges. Budgets for hiring additional, sign language trained, teachers are constrained (Makoelle & Burmistrova, 2020) The use of sign language interpreters to facilitate communication has been shown to limit direct teacher communication with the Deaf student (Cawthon, 2001). It has even been suggested that the use of an interpreter in class goes against the spirit of inclusion by isolating the Deaf student from direct contact with the teacher (Murray et al., 2018).

Thus, the goal of inclusive education for the Deaf in the Global South - the provision of accessible education in mainstream classrooms - is often reduced to minimal representations, where teachers either, use sign to support their speech, which is not direct communication what is said, and is not accessible to all Deaf learners (De Meulder et al., 2019); or they use just enough sign language to get by, but not enough to ensure equity for the Deaf learner (Swanwick Goodchild & Adami, 2022); or teachers may pick up a few key phrases, which can do more harm than good (Scott & Henner, 2021).

The Role of AI in Educating the Deaf

For more than three decades, artificial intelligence has been touted as a disruptor of traditional approaches to education (Schank & Edelson, 1989; McArthur, 1995; Seldon & Abidoye, 2018; Williamson & Eynon, 2020). Important applications of AI, such as speech recognition and synthesis (Verma et al., 2021) performance monitoring (Kumar, 2019) adaptive learning (De Melo, 2014), interactive learning environments (Ijaz, 2017) and intelligent tutoring systems (Weitekamp, 2020) all hold great promise for easing the burden on teachers, enhancing student engagement and improving learning outcomes. As computing technology and computing power have evolved, there has been an explosion in the number of AI-driven applications developed for education. Most recently, there has been an explosion of interest in AI-based text generation technologies (technically, generative language models), such as *ChatGPT* (OpenAI, 2023). Despite initial concern, educators are finding that, when

properly used, these technologies can be extremely useful in enhancing the teaching and learning experience (Zhai, 2022).

Beyond the contribution of AI to education, generally, there are specific communications technologies that have been improved by modern AI techniques, which have particular implications for Deaf education. Consider gesture recognition and analysis and machine translation, which all play a role in improving access for learners who do not benefit from traditional approaches to teaching and learning. For inclusive education, the Deaf have particular needs that AI can help to address. Firstly, in a traditional classroom, it is unlikely that the teacher will have the ability to sign. Here, tools developed using AI and deep learning can be used to allow the Deaf student to communicate his ideas to a teacher that does not sign and for the teacher to communicate seamlessly with the Deaf student. With the modern AI techniques and access to high performance computing, the role of AI in enabling equitable education for the Deaf has never been more evident or more accessible. Given the issues of equity in education of the Deaf, and the issues of transfer of this concept from the developed world, it is clear that AI is required to address the cultural issues that present themselves when Deaf students are included in mainstream classrooms (Mohammed & Watson, 2019). Here, one must consider issues of fairness in AI. The lack of representative data in training and inaccessible interfaces are issues that impact the usefulness of AI for the Deaf (Kaffle et al., 2019). Bragg et al., have considered the "bias, privacy, accuracy, and over-promotion" issues related to the promise and implementation of AI technologies for people with disabilities (2023, p. 1). Their analysis of the FATE (Fairness, Accountability, Transparency and Ethics) of AI data use, as it relates to the sign language translation, highlights the fact that it could exacerbate some of the traditional discrimination and oppression of the Deaf (Bragg et al., 2021). Additionally, there is concern that the advent of sign language translation systems has led to the ambition of replacing human signers with AI-based avatars before these systems are good enough to effectively replace human signers (Al-Khazraji et al., 2018, Kaffle et al., 2019).

There has long been a recognition of the potential for AI to positively impact Deaf education (Parton, 2006). Parton's review explored research into computer-aided instruction, sign language recognition, sign language synthesis and signing avatars. Almost 20 years later, very few of these technologies have made their way into classrooms and there is no coherent approach to using AI to enhance the teaching and learning experience for the Deaf community. This gap has been highlighted by a recent review of technologies for inclusive Deaf education in which the AI-based studies reviewed featured applications for one-way translation of speech-to-sign or sign-to-speech, as well as speech-to-text and speech synthesis solutions (Rodríguez-Correa et al., 2023). The contention of this paper is that this is insufficient, and that an AI-based system that simultaneously allows for signing by the Deaf student and speaking by the non-signing teacher is required for inclusive Deaf education in the Global South.

Previous Solutions for Deaf Education

Previous work aimed at accommodating Deaf students in classrooms with hearing students have proposed solutions that can broadly be grouped into three categories: systems that convert speech to text (Iglesias et al., 2014; Butler, Trager & Behn., 2019); systems that convert speech into a transcript that drives a signing avatar (Bouzid & Jemni, 2020; Adamo-Villani & Anasingaraju, 2017); and systems that convert sign to text (Reis et al., 2015; AlShammari et al., 2018; Bragg et al., 2019).

Iglesias et al. (2014) propose a speech-to-text system under the APEINTA project. The system produced real-time captions from the speech of the teacher, which was either projected or sent to a mobile device for the Deaf student to read. The recognition system utilised a commercial automatic speech recognition (ASR) system – Dragon Naturally Speaking – to produce captions for Spanish students. Butler, Trager & Behn (2019) implemented a captioning system, driven by ASR, as a means of providing university students with multimodal access to spoken language. The students also had access to sign language interpreters. The researchers reported good reviews for the captioning system, despite the errors of the ASR system.

Sign language avatar systems have been making significant strides over the past 40 years, due in large part to increasing computing power and, more recently, deep learning algorithms (Aziz & Othman, 2023; Rastgoo et al., 2023). Dhanjal and Singh (2022) developed a system that automatically converts speech to Indian Sign Language (ISL). The system combines a multi-language ASR with machine translation techniques, transforming the ASR transcript into HamNoSys sign language notation (Hanke, 2004). A signing avatar was then developed and deployed for testing. The resulting accuracies and acceptability ratings were favourable and the authors pointed to the value of the tool for educational as well as social settings. The reviews by Bouzid & Jemni (2020) point to a number of speech-to-avatar as well as text-to-avatar systems, using a variety of technologies, including machine learning. Many of these systems were tested in the classroom with favourable results.

An emerging trend is the use of avatars for communication with the Deaf in mixed-reality (MR) classrooms (Miller et al., 2017; Adamo-Villani & Anasingaraju, 2017). Yang, Mousas & Adamo (2022) built an MR application for use in teaching K-1 Math lessons. They used motion capture technology (OptiTrack, StretchSense, and FaceWare software) to create an avatar (using Autodesk's Motion Builder) of a professional deaf signer going through the lessons. Participants in the study (eight, hearing adults) found the appearance acceptable and the signing function, this despite the high cognitive demand of interacting with the system. The major challenge with the study was that it was not done with Deaf children in a real classroom. The study by Imashev et al. (2024) sought to overcome this with a study of 21 Deaf children in an inclusive classroom using MR to teach lessons on animals.

As with sign language production system, sign language recognition has been revolutionised since the introduction of deep learning – see Núñez-Marcos et al. (2023) for a review. However, very few applications have been developed directly for inclusive education and fewer have been tested in the classroom environment.

Papadimitriou et al. (2023) developed a web-based, Greek sign language recognition system for use in education. They employed a 3D convolutional neural network

(CNN) for isolated word recognition as well as MobileNet, a 2D-CNN for continuous fingerspelling. A bidirectional long short-term memory (BiLSTM) encoder was used to combine the separate streams. The system was tested with 23 users, a mixture of university students and system developers. Quantitative analysis of the accuracy showed that the system worked well for multi-signer models, but degraded with signer-independent recognition. The user experience was reported to be positive.

There are three major drawbacks with the speech to text approach. Firstly, these systems assume that the Deaf have a command of written language, which most do not (Volpato et al., 2018). Secondly, there are most often errors in the transcribed text (Marschark et al., 2005; Marchut, 2017). Finally, switching between sources of information (a screen and the instructor, for instance) increases the cognitive load and makes learning more challenging (Kushalnagar, 2016). Speech to avatar systems go beyond producing a written transcript, using that transcript to drive an avatar that signs. This type of system allows the Deaf to understand what was said by the teacher, but it does not allow the Deaf to sign and be understood. Similarly, sign to text systems perform gesture recognition to produce text from the student. However, if the instructor does not sign, the communication with the student is limited.

Each of these AI-based technologies is useful by themselves, but do not address, at the same time, the needs of the Deaf in an inclusive way. For instance, a sign-to-speech system will allow for the Deaf student to communicate their thoughts to the teacher, but the teacher who does not sign, will not be able to respond directly. This is where a sign language interpreter would be called upon to convey information to the student. However, recent advances in sign language avatars (see, Bouzid & Jemni, 2020 and Atasoy, 2023, for reviews) promise to address this deficit.

Despite the existence of a number of enabling technologies, there is still a need for a solution that allows for the hearing teacher and the Deaf student to communicate seamlessly. Given the myriad challenges in increasing the number of Deaf teachers, hearing teachers that sign, or sign language interpreters, in mainstream classrooms, the most practical approach is to develop closed-loop AI-based systems that will be deployed alongside the hearing teacher that will allow the teacher to speak and the student to sign and to have them understand each other using their preferred mode of communication. No such system currently exists; the system being proposed in this paper is meant to fill this gap. The following section discusses the theoretical framework that guides the development of the solution.

Theoretical Framework

This section explores the theoretical framework that informs the development of the proposed inclusive solution that aims to facilitate communication and learning for Deaf students in classrooms with hearing students and a hearing teacher, that does not sign, where no interpreter is present. One theoretical framework that informs the development of the inclusive solution is Positioning Theory. Positioning Theory facilitates an exploration of language, including sign languages and other sign systems, as dynamic, multifaceted discourses while recognising how people actively choose and change their roles in conversations, making language a lively and complex interac-

tion. It also posits that individuals construct their identities and perspectives within social contexts, including their positioning within social contexts, their relationships with others and interactions with various discourses and power structures. Positioning Theory emphasises how individuals adopt certain positions or roles in discourse. The positions are seen as fluid and changeable over time, depending on the context and the discourses. The value of Positioning Theory to the proposed system lies in the insight it provides into how individuals navigate social interactions and construct meaning in specific contexts. Specific to the development and deployment of a signing avatar for Deaf students in mainstream or inclusive education settings, Positioning Theory informs this research in promoting: (1) an understanding of identity and the positioning of Deaf students in the classroom, (2) the empowering of agency on the part of Deaf students, (3) navigation of the power dynamics of social interactions with the Deaf students and their hearing teaching and student counterparts and (4) facilitating a more inclusive education experience for the Deaf.

In addition to Positioning Theory, the theoretical framework also integrates aspects of Universal Design for Learning (UDL), Cultural Responsiveness, Vygotsky's Polyglossia, Sociocultural Interactions, and the Matching Person and Technology Model.

UDL is a framework that emphasizes designing curriculum and instructional materials to accommodate the widest range of learners possible without the need for adaptation or retrofitting (Hitchcock et al., 2002). Applying UDL principles to the development of technology for deaf students involves ensuring multiple means of representation, engagement, and expression. For instance, the technology will support various modes of communication, such as sign language and speech, to cater to diverse learning preferences and abilities and to reduce the burden on the school and classroom teacher, who would have to hire and accommodate a sign language interpreter to support the Deaf learners.

Combining cultural responsiveness with UDL – as proposed by Silvesti and Hartman (2022) – allows for the recognition and valuing of the various cultural backgrounds, identities, and experiences of students and teachers. For Deaf students, their cultural identity often revolves around sign language and Deaf culture (Andersson & Adams Lyngbäck, 2022). Therefore, the technology should not only support sign language but should also incorporate features that reflect the linguistic and cultural nuances of the deaf community. This aspect of the solution should be highlighted during training of classroom teachers, so that they can provide access to resources and content that are culturally relevant and affirming for deaf students.

Vygotsky's concept of polyglossia emphasises the coexistence of multiple languages and forms of communication within a social context (Vygotsky, 1987). In the context of developing technology for deaf students, polyglossia underscores the importance of supporting both sign language and spoken language as complementary means of communication. Even though the majority of Deaf students in the Global South, who this solution is initially aimed at, do not use spoken language, those that do will benefit from the multimodal input. The technology should facilitate dynamic interactions between sign language and spoken language, enabling seamless translation and comprehension for both the deaf student and the teacher.

Sociocultural interactions (Vygotsky, 1987) refer to the social exchanges and collaborative learning experiences that occur within a cultural context. For deaf students

in mainstream classrooms, effective communication and interaction with peers and teachers are crucial for academic and social development. The technology will allow for multiple Deaf students to be accommodated in a mainstream classroom, which should promote meaningful socio-cultural interactions by fostering inclusive communication environments where Deaf students can actively participate and engage with their peers in academic discussions, group activities, and social interactions. As currently envisioned, the tool does not allow for interactions of the Deaf with their hearing counterparts, which is a limitation that can be addressed once the first version is deployed.

The Matching Person and Technology (MPT) Model emphasises the importance of aligning technological solutions with the unique needs, preferences, and abilities of individual users (Martinez, 2018). When designing technology for Deaf students, it is essential to consider factors such as language proficiency, communication preferences, sensory capabilities, and educational goals. The technology should be customisable and adaptable to accommodate the diverse needs and language preferences of Deaf students, empowering them to fully participate and succeed in mainstream educational settings.

By integrating the principles of the various theoretical models, a tool should emerge that considers every aspect of inclusive Deaf education. This comprehensive approach to the development of technology for Deaf students in mainstream classrooms, it is thought that the barriers of sign language communication can be removed and that a more inclusive learning environment can be fostered, to enhance educational opportunities and empower Deaf students to thrive academically and socially.

A Proposal for AI-Enabled Inclusive Deaf Education

Given the reluctance of some in the Deaf community to consider inclusive education that is not culturally sensitive or accessible, an equitable solution has to be envisioned. The solution must allow the Deaf person to engage in learning using their native language and not require them to learn to lip-read, or to demand that they have a command of the majority language of their country.

The proposed approach addresses shortcomings in these systems, by facilitating real-time two-way communication between Deaf students and non-signing teachers. In order to achieve this, the approach combines speech and language technologies, computer vision, machine translation and 3-d avatars. The most significant difference between the proposed approach and others is that none of those currently in existence advance a closed loop system (see Sect. 2.4, for a discussion). That is, they do not allow for real time communication between the hearing and the Deaf. This innovation will make the proposed system the first of its kind. Acceptance of the system, by both the Deaf and the hearing, is more likely, as it allows them to interact using the mode of communication with which each is most comfortable. One major drawback would be the realism of the signs produced by the avatar (Coy, 2022). This system would have to be able to interpret the non-verbal features of communication that impact meaning and convey such meaning in the signs produced by the avatar. Machine translation between spoken and sign language is another potential challenge

of the project. Unlike machine translation between spoken languages, the machine translation from sign language to spoken language would need to interpret the non-manual markers of the sign utterances of students and convey the appropriate meaning in the translated utterance.

The proposed system will allow for the integration of Deaf students in main-stream classrooms with a teacher that is not competent in sign language. This will allow for improved education and increased employment opportunities for Deaf students, who currently have few prospects for intellectual development or social advancement. Also, exposing hearing individuals to the Deaf could change attitudes to the Deaf. It is anticipated that the main output of this work will be a system that allows a Deaf child to be able to attend the school of her choice all the way through to university without the need for expensive training for teachers and support staff or the requirement for Deaf students to adapt to learning in English. This is true equity. What currently obtains in much of the developing world is that the Deaf attend specialist schools with little technological innovation, no curriculum specific to Deaf education, and no structured pathway to higher education. The proposed system will offer the Deaf a choice of schools, where they can learn in their own language and will have access to the type of education that is suitable for their chosen career path.

speak to how the theoretical framework fits with the system structure:

The proposed system would make it feasible for the voice of the Deaf to be heard and for their linguistic and cultural positioning in the classroom to be validated. When a human interpreter is in the classroom, the Deaf student has to rely on them to ask for clarification, request feedback, and express a breakdown in comprehension. In these instances, the Deaf students may fail to experience self-agency. Positioning Theory acknowledges the influence of power dynamics on social interactions and therefore the deployment of a signing avatar empowers the Deaf student to actively and instantly enagage in communication and learning despite the absence of the ideal- a signing educator. In the context of hearing teachers and peers interacting during teaching and learning activities and especially those featuring collaborative processes, the use of a signing avatar helps to mitigate the power imbalance by facilitaing equitable access to communication and the learning resources. Deaf students would now have to opportunity for fuller participation in classroom discourse and the dependency syndrome would be stymied. By leveraging Positioning Theory through the use of the signing avatar, this solution promotes inclusive practices and accommodates this crucial diverse need that revolves around the language deprivation issue that defines communication experiences of the Deaf. Indeed the application of Positioning Theory to the development and deployment of the signing avatar for Deaf students in school placements where their teachers and peers are both hearing and do not sign, heralds the value of recognizing and validating students' identities, empowering their sense of agency, navigating the power dynamics during classroom discourse and promoting inclusive education practices.

System Structure

The objective of this research is to design and implement a system that will facilitate real-time, direct communication between the Deaf and hearing in the classroom.

In order for the Deaf to understand the hearing, the system will accept real-time, continuous conversational speech as input and produce sign language, signed by an animated avatar. When the Deaf person signs, it is recognised by the system which produces an output of spoken English so that the hearing individual can understand what was signed. Figure 1 outlines the main steps involved in the proposed solution, which can be deployed directly in the Global South. Given the modularity of the system, specific modifications could make it usable in the Global North. An ASR system capable of accurate real-time speech recognition will be developed. Such a system is possible, given recent advances in end-to-end, low-latency ASR systems that work in real-time, even for low-resource languages (Kannan et al., 2019).

A machine translation system capable of translating spoken language to sign language and a computer animated avatar that signs are the other novel components of the Speech to Sign Language section of the project. Recent advances have been made in the development of speech to sign language avatar systems for real-time speech to signing avatar systems for use in real world applications (Kumar et al., 2020; Waldow & Fuhrmann, 2020). For the Sign Language to Speech section, existing (or novel) gesture recognition and machine translation approaches will be incorporated.

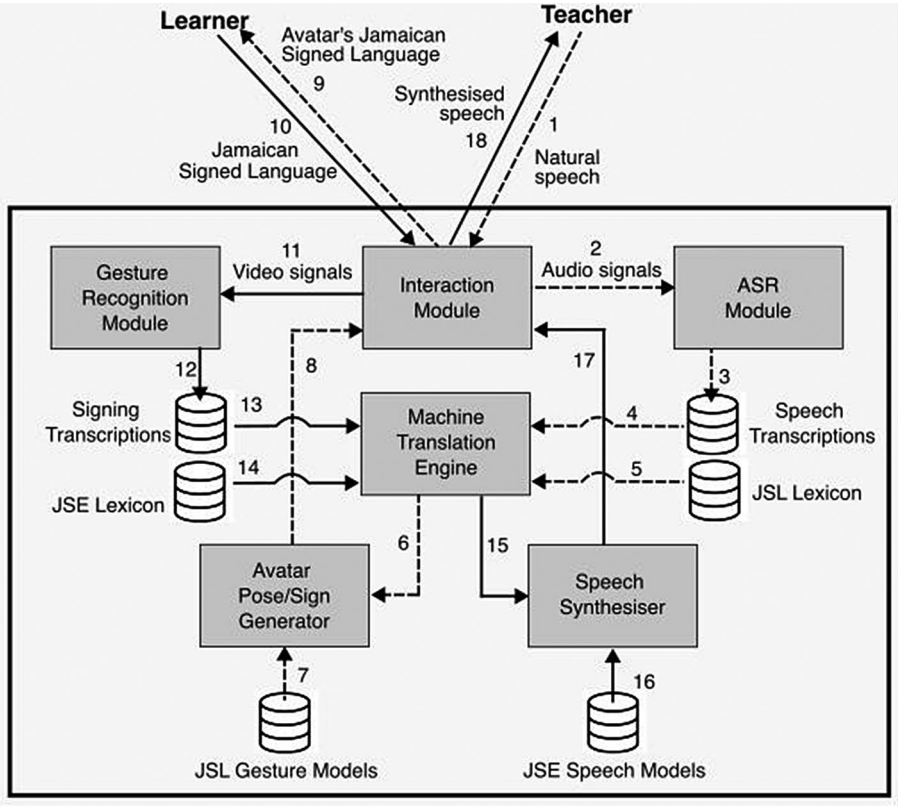


Fig. 1 An overview of process flow of the proposed closed-loop solution using, as an example, Jamaican Sign Language (JSL) and Jamaican Standard English (JSE). The modularity of the solution will allow for any pair of languages to be used

Figure 1 indicates the use of Jamaican Sign Language (JSL) and Jamaican Standard English (JSE) as the target languages, however, the system is meant to be modular and any pair of languages can be used, depending on the domain in which the tool is to be deployed.

Sign language recognition is a maturing field, with a range of techniques being developed, from generative hidden Markov models, to the more recent deep learning gesture recognition techniques (Bragg et al., 2019; Pu et al., 2019; Bantupalli & Xie 2019). Novel sign language translators are being explored that treat the recognition and translation of sign language as an end-to-end process, accomplished using deep learning approaches (Bantupalli & Ying, 2018; Camgoz et al., 2019; Guo et al., 2019; Yutong, 2022). Modern end-to-end synthesis approaches make the development of realistic synthetic voices accessible, even for low-resource languages; these techniques will be employed to model speech (Fu et al., 2020; Zhang & Lin, 2020; Mohammed & Coy, 2021).

A Deaf Perspective on the Solution

Participants

In an effort to gauge the willingness of members to accept an AI-based closed-loop solution to the challenges of educating the Deaf in Jamaica, a focus group was held in March 2020. The session was conducted at the weekly Deaf Club meeting in Kingston, Jamaica with 14 of the leaders in the Deaf community (five females and nine males, all of whom were in the age range 21–60), one hearing individual – a graduate student, who is also a trained interpreter – and an interpreter selected by the members of the Deaf community. Six of the participants were tertiary educated – one was enrolled in university at the time of the meeting. The others were educated to the secondary level. None of the participants had an assistive device.

Sampling

Convenience sampling was employed, as the focus group was held during the height of the COVID-19 pandemic. Recruiting for the focus group was done indirectly, through the Jamaican Association for the Deaf (JAD). Owing to the closed nature of the Deaf community in Jamaica, the JAD was approached to act as a go-between and to seek an invitation for the researcher to attend a meeting of the Deaf Club to present the idea of the closed-loop system to the attendees and collect their feedback. Thus, the sample consisted of the members of the club who attended the meeting on that night. Additional sessions were planned with other members of the club, but the country went into a series of restrictions, including lock downs, which prevented meetings of the Deaf Club. Given the exploratory nature of the study, the small sample size was deemed to be justified, as there was no intention to generalise the resulting responses to the wider Deaf community.

Table 1 Guiding questions for Focus Group

1. Did you use technology for education and was your experience with technology a positive one?
2. How would you feel about having a device in the classroom that would replace an interpreter?
3. Is there a general reluctance among the Deaf community to use technology in the classroom?
4. How would you feel about a software tool that allows for two-way communication between a hearing teacher and a deaf student in a mainstream classroom?

Methodology

The session was facilitated by the first author with the assistance of the interpreter. The facilitator stood at the front of the room and spoke orally, while the interpreter signed the presentation of the facilitator to the Deaf participants. When the Deaf participants signed their feedback, questions and comments, the interpreter spoke orally, so that the facilitator could hear and respond. The facilitator's responses were then signed to the Deaf participants. Video and audio recordings of the session were made and transcriptions were produced after the session for future analysis. The session was divided into two parts; it began with engagement on the use of technology for education and the barriers to technology use in Deaf education (Coy, 2022). Next, the participants' views on the use of technology for Deaf education were elicited. As a part of this discussion, the facilitator introduced the notion of the closed-loop system described in Sect. 4 and then played a video of a signing avatar[2] for the participants, as an example of what a signing avatar could do. The format of the focus group session was to have an open discussion with the participants and to only intervene when clarification was requested. Guiding questions were used to stimulate discussion (see Table 1).

The resulting exchanges were analysed for themes and sentiment. ChatGPT-3.5 (OpenAI, 2023) was employed for the thematic analysis - the results of which were verified by all three authors – while sentiment analysis was performed using a transformer-based sentiment analysis toolkit, the *pysentimiento* toolbox (Pérez et al., 2021). For the thematic analysis, a transcript of the session was generated. The authors worked independently to identify the key themes by reading through the transcript and extracting sections that followed a coherent theme. The separate analyses were shared and overlapping themes identified. Where themes could be merged, a decision was taken to do so. Additionally, the sections of the transcript that contained the utterances of the Deaf participants were passed to ChatGPT3.5 and a prompt to generate a thematic analysis was given. The resulting themes aligned with those determined by the independent analyses of the authors.

[2] The demo was one of Dutch Sign Language using an avatar developed using JASigning: https://vhg.cmp.uea.ac.uk/demo/movie/WelkomNGT.mov.

Results

Thematic Analysis

Five main themes were extracted from the transcript. The themes were: avatar effectiveness and realism, resource and cost concerns, language and cultural considerations, real-time translation challenges and community involvement and feedback.

Avatar Effectiveness and Realism

The participants had two major concerns regarding the potential effectiveness and realism of the avatar. Firstly, they mentioned that it was not human-like in its appearance.

> *… and also the color of the complexion is just poor. And it seems like it's a robot for real. It's not human-like. I wasn't impressed of the [sic] complexion of the person when the person was signing.* [Participant 3]

Secondly, the Deaf participants of the focus group had real concerns about the potential of the avatar to effectively represent their language. Here, the participants had two primary concerns, the multimodal nature of signing and whether the avatar would be effective in situations where a Deaf relay interpreter would be required. The participants made the point that accurate signing involves facial expressions as well as gestures.

> *That's not effective. Let me explain why I say that. For example,*
> *prefix like prefix are some other words where you can add… UN.*
> *For example, you have different form of prefix[sic]. We don't have a sign for those.*
> *prefix. It's how we express, by using our facial expression. So how would you do.*
> *that with the avatar when it comes on to the prefix part of our language?*
> [Participant 6]

Deaf relay interpreters (DRIs) are experienced Deaf interpreters that provide brokering services to other Deaf individuals (Adam et al., 2014). Their roles vary, but they often provide support for Deaf signers with mental challenges, home signs or vision challenges. In some instances, they work alongside hearing interpreters to assist the Deaf where unfamiliar concepts are being discussed, and where the sign being used by the hearing interpreter needs clarification. Given that the contexts within which signing avatars will be used (education, healthcare, transportation, etc.) may require clarification of some concepts, a system that employs an avatar for communication with the Deaf should consider how these clarifications will be provided.

> *… I have two things. It comes with interpreters and DRI. Deaf relay interpreters. For interpreter purposes, you have to sign naturally and sometimes persons*

don't understand. So that's why we need a deaf relay interpreter in between the hearing interpreter and the deaf person.… Possibly a deaf individual won't understand. For example, a deaf English word comes down to different words. A deaf person may understand that. For example, idiom.… How will the system incorporate those idioms, those prefix and so forth into the sentence, into Jamaican Sign Language? Because in English it is very easy. [Participant 4]

Language and Cultural Considerations

The participants expressed concerns about the avatar's ability to understand and translate multiple languages and the importance of cultural and linguistic sensitivity. This concern arose from a discussion of the assumption that the teacher in the classroom would be using Jamaican Standard English for instruction.

So my concern, suppose the teacher is speaking patois with your program or whatever, you are able to understand clearly and translate it to JSL. [Participant 4]
So will your program be able to understand the three languages, English, patois, and JSL? [Participant 5]

This thought led to the questioning of the motives for developing the avatar, as it related to Deaf culture. One participant asked if the avatar was being developed to replace interpreters – a common concern among the Deaf (European Union of the Deaf, 2023; World Federation of the Deaf, 2023, Kipp et al., 2011).

I'm just curious, what's the reason behind the development of that avatar? Do you have a problem with interpreters or what's the reason behind the development of the avatar? [Participant 12]

Another significant concern expressed, was that hearing parents of Deaf children may not find it necessary to learn sign language, if they think that a signing avatar can be used for real-time communication. This approach, they thought, would lead to hearing parents not interacting directly with their Deaf children.

And for young children, I think this system can be a bit negative for them because their parents don't learn to use sign language to communicate with them one-on-one. [Participant 3]

Resource and Cost Concerns

The participants wanted to know if the system would come at a cost to them. They recognised that there would be a cost to develop the system and to provide the classroom-based infrastructure required for successful deployment. Their concern was that the Deaf students, or their parents would have to pay a premium for access to

the system. There was fear that, as they had experienced before, the Deaf community would be used as a way to get project funding, which was misappropriated and the community received no benefit from the project. Participants were keen to ensure that cost would not be a barrier to access for Deaf children.

> *… will that app be free?…*, *when you develop an idea for the deaf community*, *it's fantastic. I like it. But most times*, *the deaf person feels robbed and feel neglected and the person benefits from us. So*, *they go out*, *they start project and benefit from us*, *but we as deaf people don't benefit at all. So*, *they use the deaf community as a name to get funds for themselves. So… And that is not right. So*, *the deaf community*, *most times*, *they write to our views. I understand. So*, *before you finalise the development*, *make sure that the deaf club can have full access.* [Participant 1]

One participant expressed the view that the money would be better spent on training interpreters, or training hearing teachers in sign language, who could be deployed in the various schools. A discussion ensued about the comparative benefit of spending on system development as opposed to interpreter training. Most participants seemed to accept the logic, if not the substance, of the argument.

> *Okay*, *no problem. We're talking about costs when it comes to employing interpreters. That is true. And we have to save on costs. I agree. But we can save*, *we can spend costs on other things. For example*, *when it comes on to the hearing school from primary to high school*, *they must learn how to sign. That would be good. All the teachers. So*, *in a deaf school*, *teachers aren't skillful in those areas. When it comes on to the hearing school*, *the teachers are good in speaking and communicating. But for the deaf individual*, *they don't have so many skilled teachers. Teachers have their degree in primary school. So*, *I think when it comes on to our native language*, *it should be invested in that area and qualified teachers to communicate with us as deaf individuals.* [Participant 5]

Real-time Translation Challenges

The technical challenges were raised by the participants; they had concerns regarding real-time translation and its impact on classroom interactions and learning experiences. The proposed approach consists of a number of integrated sub-systems working together to facilitate classroom-based communication between Deaf students and hearing teachers. It was pointed out that each of the sub-systems would have an associated time lag. When combined, they feared, the total system lag would hinder real-time participation of the Deaf student.

> *When the teacher is talking in… English*, *then translate[s] to Jamaican sign language… So it will be five minutes lagging*, *ten minutes*, *one minute*, *15 s because that would be a problem because when the teacher talks*, *the hearing person*, *they get it quick quick but for the deaf persons to benefit from what the*

teacher says. It depends on the waiting time for that, for the voice to be translated to text and then to sign language, so what's the lag time? [Participant 3]

Participants pointed to other fora, such as plays, where interpreters were used. Often, the lag in signing caused them to be behind the rest of the audience, getting the punch line when the hearing audience had already moved on. They pointed out that in an educational setting, the student would have bigger issues than the mild annoyance they feel at the theatre; they made it clear that there could be real harm, frustration and disengagement.

The same challenge with questions for example deaf and hearing are in the class the teacher is there talking… the hearing students always answer first because the deaf person is always lagging behind when it comes on to a physical interpreter. So when the deaf person stand[s] up to sign or to ask a question. The deaf person understand[s] [and] answer[s] same time and the hearing person is lagging behind so it's the same thing when it comes on to the two parties involved. [When the] Hearing [person is] talking, the deaf person is lagging behind and [when] the deaf person is signing and the hearing person is logging behind. When it comes on to hearing they're talking to the mic and when they speak the hearing person claps and the deaf have to wait until the interpreter reaches that applause phase and then they give a round of applause. So it's always lagging behind. [Participant 3]

Community Involvement and Feedback

Finally, participants emphasised the importance of involving the Deaf community in the development process.

Will it constantly involve the community in the development of the project? [Participant 4]

Sentiment Analysis

Sentiment analysis was performed using the *pysentimiento* toolbox (Pérez et al., 2021). The sentiment analysis tool is trained with the *SemEval 2017* corpus (Rosenthal et al., 2019) using a pretrained RoBERTa model, BERTweet, which is trained on 850 million English tweets (Nguyen et al., 2020). The text of the participants' statements was tokenised into units of topical coherence, to ensure that part of a thought was not separated into unrelated chunks. These were then input to the sentiment analyser, which output one of three labels, 'POS', 'NEU' and 'NEG', indicating whether the input was positive, neutral or negative in sentiment. Along with the label, a percentage certainty was output by the tool. If the percentage certainty fell below 80%, the response was manually examined and the label corrected, if it was erroneous.

Figure 2 shows the result of the sentiment analysis. Almost 50% (49.61) of the statements were neutral, while 45.76% and 4.6% were negative and positive, respectively.

Discussion

Thematic Analysis

The focus group results raised many issues related to the design of a closed-loop system for inclusive education of the Deaf. One of the most striking is the issue of cultural responsiveness. The appearance of the avatar proved to be an important concern for the participants. This is a common concern for the Deaf when they are introduced to the use of avatars for signing (Quandt et al., 2022) – Kipp et al. (2011) found that the Deaf thought naturalness to be the second most important feature of an avatar. It has been shown that the Deaf showed a preference for natural-looking avatars over more cartoonish anime-style avatars (Brock et al., 2018). However, there seem to be limits to this preference. Adamo-Villani et al. (2015) found that a realistic, human-like avatar was less acceptable to users than a stylised one – as pointed out by Quandt et al. (2022) this preference may be due to the uncanny valley effect (Mori et al., 2012; Shin et al., 2019; Wolfe et al., 2021). Users can, therefore, be significantly impacted by the appearance of an avatar; if the impression is unfavourable, then the interest in the avatar will be low, even if the signing is intelligible (Adamo-Villani et al., 2016). This may even lead to an overall rejection of the avatar (Naert et al., 2020). This speaks to the need for cultural responsiveness in the design. The requirement

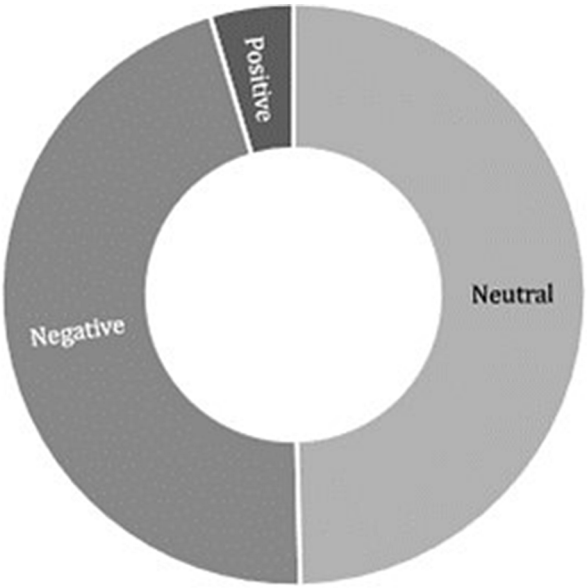


Fig. 2 Sentiment analysis of the deaf participants' statements

that the avatar be representative of the population resonates with the issue of representation and potential bias in AI-based systems for the Deaf (Kaffle et al., 2019).

A related concern is that the avatar should effectively represent their language - This issue has been one of the most challenging aspects of signing avatar development (Wolfe et al., 2021). Despite the challenges, significant progress has been made in recent years to improve the synthesis of the non-manual elements of signing (Wolfe & McDonald, 2021; Johnson, 2022). Additionally, there are novel approaches to facial animation that have potential application in the development of signing avatars (Hussen et al., 2021; Laine et al., 2021; Vougioukas, 2019).

These concerns aren't simply cosmetic, they speak to the very heart of Deaf culture. Many Deaf individuals define themselves by their language (Andersson & Adams Lyngbäck, 2022) thus any inappropriate representation of it is an affront to them. This is where UDL comes into play in the design process. As posited by Silvesti and Hartman (2022) a culturally responsive design of the proposed closed-loop system must respect the sociocultural backgrounds, identities (linguistic and cultural) and preferences of the Deaf student, if a desirable outcome is to be achieved.

The issue of Deaf Relay Interpreters, to the best of the authors' knowledge, is an area that has not been discussed in the literature. It is possible that this has been overlooked because the focus has been on generating an avatar to produce accurate, natural-looking signs and less on the deployment of that avatar in a real-world application. As the technology improves and more avatar-led communications systems are developed, attention should be given to how the role of the Deaf Relay Interpreter will be replicated, as stated in World Federation of the Deaf and the World Association of Sign Language Interpreters statement "Translators and interpreters take time to prepare in order to fully understand a message and to give consideration to their target audience. An automated translation will not be able to take other sociolinguistic and sociocultural factors into account in order to ensure that the intent of the message sender, and the goals of the message receiver, are achieved." (World Federation of the Deaf, 2023, p. 1). This is where an AI-based system could be developed to model the appropriate sociolinguistic and sociocultural elements of deaf relay interpreting in order to enhance the utility of the proposed system for a wider range of Deaf learners.

When the participants pointed out the need for supporting bilingual Jamaican speakers, it brought up the relevance of Vygotsky's concept of polyglossia in the system design in an unexpected way. The idea of support for multiple languages, in the context of inclusive Deaf education, is usually focused on providing language support for the Deaf who have access to both spoken and signed language. The fact that Deaf individuals are aware of the language needs of hearing teachers of the Deaf is a revelation. The Jamaican context is one in which most speakers are either monolingual speakers of the local English-based creole (referred to as Jamaican by linguists, but colloquially called Patois) or bilingual Jamaican/English speakers (Farquharson, 2013). While English is the official language of instruction, it has been found that most teachers selectively use Jamaican in the classroom (Nero, 2022). This raises the issue of the ASR system being able to effectively recognise what the teacher says – the hearing students would understand, but the avatar would not be able to sign it. This, the Deaf participants, linked to the culture of the teacher and the apparent effort to stifle the culture, by limiting the teacher to speaking one language over the

other. This issue is distinctly a problem in the Global South, as creoles were developed in post-colonial societies, which is also where their use is still mostly seen as problematic (Krämer et al., 2022). Thus, the MPT aspect of the framework becomes relevant to both the hearing and Deaf users of the system – an aspect of the design that can be facilitated by the use of AI-based technological advancements in speech and language technologies.

Participants were in general agreement that it would be better for all teachers in mainstream classrooms to be taught sign language, so as to accommodate Deaf learners – this is echoed by some in the literature (Skyer, 2020). The rationale given was that there would then be enough teachers with the language skill to help the Deaf directly in the classroom and there would be no need for a signing avatar. The primary concern being expressed here, seems to be the fear of cultural marginalisation, which developers of signing avatars have to take seriously and address with the community (Wolfe et al., 2021). However, even the staunchest advocate for signing (even Deaf-only) teachers of the Deaf recognise that the concept is somewhat idealistic (Scott et al., 2023). While the reception to signing avatars may not be ideal, based on the view that signing humans, the logistics and cost implications (Gamal, 2022; De Meulder & Haualand, 2021) suggest that the best option for inclusive Deaf education in the Global South is to have a system that allows for the use of signing avatars.

Despite the potential value of having a signing avatar, participants were sceptical. In particular, they felt that hearing parents of Deaf children may not find it necessary to learn sign language, if they think that a signing avatar can be used for real-time communication. This would be quote harmful, given the importance of early sign language learning to the overall development of the Deaf individual (Lieberman et al., 2024). This is a similar concern to that expressed about the rise in popularity of the cochlear implant (Wolfe et al., 2021). This cannot be ignored, but it should not be a barrier to technological innovation in inclusive education. One approach being taken, is to provide early interventions and support for hearing parents of Deaf children to learn sign language (Humphries et al., 2019; Terry & Rance, 2023; Lieberman et al., 2024).

The cost implication of inclusive education is significant for countries in the Global South (Coy, 2022). These concerns came out strongly in the focus group session. So much so, that past experiences of exploitation of the Deaf came up. It was indicated that, in the past, the Deaf community has been taken advantage of by individuals who got project funding and did not deliver, while engaging the deaf, raising their hopes, using them to gather information and ultimately, wasting their time. This non-tangible cost was seen as significant and weighed heavily on their minds. The cost to use the system was also a question that was raised – one participant suggested it might be better to spend the money that might be used for system development, on training new interpreters. The cost of assistive technology to facilitate inclusive education has always been a concern (Borg & Östergren, 2015) especially in the Global South (Lynch et al., 2024). This is an important consideration for developers and policy makers who would have to ensure that any system developed using the proposed framework, is accessible, in cost terms, to the Deaf.

The focus group participants were sceptical of the speed and efficiency of the proposed closed-loop system. While advances have been made in the various com-

ponents of the proposed system (Kannan et al., 2019; Kumar et al., 2020; Waldow & Fuhrmann, 2020; Wolfe et al., 2022) it is certain that much work will have to be done to ensure a seamless experience for all users.

Community involvement was highlighted as being important – the Deaf participants wanted to ensure that they were integrally involved in the development of the proposed system. Previous work aimed at the community did not include them in the design process, which led to a feeling of distrust of any solution claiming to be for the Deaf (Coy, 2022). This is not a new phenomenon, as other researchers have found the same limitation in systems aimed at inclusive education (Gersten and Edyburn, 2007; Stinson, 2010). The partnership model should see Deaf individuals brought into the process at the level of project conception, as their unique perspective can only make intended systems better. Wolfe et al., (2021) go even further, to encourage hearing researchers to get involved in the Deaf community, immersing themselves in the language and culture, as much as possible, so that they gain perspective on the needs, concerns and expectations of the Deaf. This will go some way in developing trust in the developers and ultimately the solution.

Sentiment Analysis

The positive statements highlighted the participants appreciation for what they saw as the inherent value of the proposed tool and its potential application in domains other than education:

> *"I was thinking this solution is a very good proposal and I appreciate it so much. And knowing that you all want to support the deaf community, I really appreciate it. But [also for] personal use, I think it will help me. Not only [for] education, but when you go out for business purposes and other business.* [Participant 5]

The neutral statements were broadly indicative of interest, encouragement and caution:

> *I'm not stopping you working. Please continue with it. Because maybe you'll surprise us with it. So, do your thing and we will see what happens in the future? And if you're successful with the signing at all, we will let you know. We can update you and see what's going to happen. We can help you to be successful in the system.* [Participant 3]

Here the participant is expressing a certain level of scepticism, while at the same time, leaving room for the outcome to be positive and offering to help. The majority of the neutral statements took this form, identification of some value and sharing some concern. This is indicative of a willingness to give the approach a try.

Almost half of the comments were negative in nature. Some were openly dismissive:

> *But for the school students, I think it can be of assistance, but for the young ones, I don't think so. Nothing beats being natural, the interaction between the two humans. So, I don't see how it will be of any great assistance to the deaf community.* [Participant 5]

Other statements reveal issues with how the Deaf have been treated in the past, when others have proposed solutions for helping the Deaf:

> *You have noticed, but first, when you develop an idea for the deaf community, it's fantastic. I like it. But most times, the deaf person feels robbed and feel neglected and the person benefits from us. So, they go out, they start [a] project and benefit from us, but we as deaf people don't benefit at all. So, they use the deaf community as a name to get funds for themselves. So… And that is not right. So, the deaf community, most times, their rights are abused.* [Participant 1]

The lesson here is that the trust deficit can hamper acceptance of any solution, regardless of its merits. Thus, working with the Deaf community from conception to completion is not only recommended, but critical to success.

Limitations, Challenges and Possible Solutions

While there are obvious benefits to be gained from the proposed system, there are also potential pitfalls that need to be outlined and addressed. Firstly, the Deaf are often wary of technological solutions being proposed for their education, as they often take the view that hearing teachers should simply learn sign language, or that the school should provide interpreters. With the use of signing avatars, this reluctance to engage with technology-based solutions may be exacerbated, given some of the concerns expressed in the focus group. An avatar could be seen as a translator, of sorts, because it would be taking input directly from the teacher and presenting it to the Deaf student in their own language. This approach would be more feasible in mainstream classrooms, where the attempt by the teacher to sign and speak at the same time would be immensely challenging and would likely disadvantage both hearing and Deaf students. However, as pointed out in Sect. 7.1, there is recent evidence that signing avatars are seen as acceptable to the Deaf, though this is heavily dependent on the look of the avatar and the naturalness of the signing (Quandt et al., 2022). Another important finding of the study was that users who had been signing for a long time were less forgiving of the perceived unnaturalness of the avatar's signing, while younger users, and those who acquired sign language later in life, were more forgiving. This has implications for where signing avatars would be most effectively employed, with the evidence pointing to younger users being, potentially, the greatest beneficiaries. This chimes with the use of the proposed solution in the K-12 classroom.

Secondly, the technology would have to work well enough to ensure that it does not harm the Deaf learner. A hearing learner in a mainstream classroom that is exposed

to a teacher that occasionally has to self-correct, or clarify particular utterances, does not suffer harm from these interactions. However, a Deaf learner, using technology of the kind proposed here, could be significantly impacted by the cascading effect of errors in one part of the framework. For instance, an error in the speech recognition system will produce an incorrect transcript, which will be translated into sign language. The sign produced will likely not make sense to the Deaf learner. Given that error propagation, in this instance, is not linear, the Deaf learner will likely not be able to decode the message being conveyed. Similarly, the errors made in sign language recognition could produce confusing output for the non-signing teacher. The frustration of the teacher and student could have damaging effects. If the teacher is reluctant to interact with the Deaf student because the system is producing outputs that the teacher cannot understand, then that could leave the student feeling isolated and could lead them to question their place in the classroom. In fact, it could lead the student to think that there is something wrong with their signing, or even themselves - this could have knock-on effects on esteem and self-confidence.

Thirdly, there would have to be buy-in from the mainstream schools and the education authorities in the jurisdiction, otherwise, the solution would not be well used, even if it is deployed in schools. One of the most significant barriers to technology integration in schools is the lack of buy-in from stakeholders. As with all technology integration projects, there has to be a hybrid approach to both planning and execution. To facilitate top-down strategies, policy-makers and school administrators have to be convinced of the value of the project - both financially and pedagogically. They will have to be assured that the technology and requisite adjustments in the classroom will seamlessly integrate into existing instructional strategies. If these stakeholders can be convinced that the disruption will be limited and lead to improved learning outcomes, student engagement and teacher satisfaction, then they can be expected to promote its use and provide the support needed for successful implementation. From the bottom-up perspective, teachers need to be fully informed of the system's capabilities and requirements. Teacher belief, in the utility of the tool, as well as in their own ability to use it, must be addressed through multiple interactive workshops and training sessions. Beyond the initial engagement, continued support has to be given to teachers trained in the use of the technology and new provision has to be made to onboard new teachers, who would not have received training in the initial cohort.

Importantly, the solution has to be developed with the input from the Deaf at all stages of development. When the Deaf are consulted and trust the design and development process, they are far more likely to engage with the final product. Too often assistive technology solutions are developed with less than desirable consultation of end users. This is particularly problematic in the case of the Deaf, given the myriad cultural issues that are linked to the acceptance of technological aids. Early consultation will ensure that the Deaf are informed about the benefits and can express their own reservations. Further involvement in the development stage will allow for the Deaf to determine if their concerns are being taken into account and will also allow them to make suggestions for the inclusion of functionality that someone outside the community and lacking the experience of the Deaf would not have considered. This level of engagement is likely to streamline development and forego expensive remedial development and avoid loss of interest and trust among all stakeholders.

Conclusion

This paper has described an AI-based solution to the problem of lack of equity in education for Deaf students in the Global South. It outlined the issues that act as barriers to true equity, which goes beyond physical inclusion in mainstream classrooms. The proposed solution is to design and implement a system to facilitate the seamless inclusion of Deaf students in classrooms where they are taught alongside hearing individuals, by teachers with no training or experience in sign language and without the need for a human interpreter. The Deaf student will be able to communicate exclusively using sign language, which will lower one particular barrier, that of cultural sensitivity. The system will accept real-time, continuous conversational speech as input, from the teacher, and produce an output of sign language, signed by an animated avatar. When the Deaf person signs, it is recognised by the system which produces an output of spoken English so that the teacher and class can understand what was signed.

A focus group, held with members of the Deaf community, provided insights into the potential acceptance of the solution by the community. The Deaf participants pointed to avatar effectiveness and realism, resource and cost concerns, language and cultural considerations, real-time translation challenges and community involvement and feedback as being the most important concerns. Despite a fair amount of scepticism about the potential effectiveness of the system (mostly centred on the signing avatar) the participants were intrigued by the prospect of having a tool to assist with their education and were, in general, supportive of the proposed tool. The focus group revealed concerns about some issues that are likely to be more prominent in the Global South, those of post-colonial language use and socio-cultural exclusion.

Notwithstanding the limitation of a small sample size, the results of the focus group session have yielded important lessons that do not appear to have been addressed in the literature before. While it is well understood that culturally responsive design is important for Deaf users of technology, it has been revealed that it is also important, in the context of the Global South, for the hearing teacher. The focus group revealed that a bilingual-bicultural approach to developing speech recognition technologies for teachers in the Global South will need to be undertaken in order to make the system more effective.

Additionally, the concern that direct translation of spoken language to sign language may not relay the nuances of unfamiliar concepts to the Deaf learner. When the Deaf have access to a live Deaf Relay Interpreter, they may be furnished with the details they need to ensure their understanding. However, sign language production systems are not yet capable to providing this level of support. This leads to the need for further research and development of the technology to ensure that the Deaf are facilitated by the proposed closed-loop system, in the same way they would be if they had access to a Deaf Relay Interpreter.

This paper is the first step to making inclusive Deaf education a reality in the Global South. The work to be done has been envisioned and a path to a solution has been outlined. Supported by a solid theoretical framework, the proposed system is designed to overcome the shortcomings of current AI-based (and non-AI-based) technological systems deployed in inclusive classrooms. The discussions have shown

that the approach to inclusive education in the Global North is not suitable for use in the Global South and calls for a novel approach, such as the one proposed. The closed-loop system has significant advantages over the one-way systems now in place, especially for learners in the Global South, where access to spoken language is limited in the Deaf community. The vision is to take advantage of the proposed framework to broaden the prospects for educating the Deaf in the Global South in a way that values their sociocultural identity and makes learning accessible to them in way that does not require them to fundamentally alter their culture or seek access to spoken language.

In order to bring this vision, or something similar to it, there are a number of challenges to overcome. Some are technical, while others are cultural and societal. Nonetheless, there is great potential for developing a jointly agreed solution to educating the Deaf in the Global South. Given the novelty and multi-disciplinary nature of the solution, it will require collaboration across multiple disciplines and stakeholder groups to achieve the goal of improving the life chances of many of the most marginalised individuals in society. This is a call to researchers and system developers to come together to make this important step toward truly inclusive education for the Deaf.

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